



OPENNOVA



# No cloud. Your Novabot, fully yours.

The Novabot company is gone; your mower is not. OpenNova custom firmware makes it run entirely at home: no manufacturer cloud, no servers abroad, no account. It finds your own server, lets you edit the map from your chair, cuts the edges, streams its camera and repairs its own Wi-Fi. Built by owners, free and open.

100% LOCAL, NO CLOUD

OPEN SOURCE

FLASHED OVER THE AIR

REVERSIBLE

# Your mower, your network. Nothing leaves the house.

Out of the box a Novabot calls the manufacturer's cloud at [lfibot.com](https://lfibot.com). OpenNova can catch that with a DNS redirect, and for everyday mowing that is enough. Custom firmware removes the question altogether: the mower talks to your own OpenNova server and to nothing else, and the extra services on it unlock everything the factory software kept locked away.



## No cloud

Not the Novabot cloud, not anyone else's. Nothing to shut down, nothing to go bankrupt.



## Stays at home

Maps, camera, work history and your garden's location live on your own server, on your own network.



## No account

No sign-up, no subscription, no servers abroad. Your mower answers to you.



## Finds your server by itself

The mower looks for your OpenNova server on the network and follows it when you move it to another machine. No DNS tricks, no re-pairing.



## Edge cut

A dedicated pass along the border of a zone, so the last strip at hedges and paths gets cut too. Started from the app or the dashboard.



## Edit zones on the map

Draw, move and delete zones, obstacles and channels in the dashboard, and send them to the mower. No walking the mower around again.



## More than five zones

The factory firmware refuses zone six and up. Custom firmware selects zones by name, so large gardens are not capped.



## Channels the mower actually uses

Zone transits and the way home follow the channels you recorded, instead of the shortest line through the flower bed.



## No cloud, not even as a fallback

The factory firmware is built around the manufacturer's cloud addresses. Custom firmware points the mower at your own server from the first boot, so there is nothing to redirect and nothing to fall back to.



## Live camera

A video stream from the mower's own camera, in the app and the dashboard. Headlights on and off from the same screen.



## Joystick with blades

Drive the mower by hand, blades on or off, straight from your phone.



Your garden as the dashboard can show it: a 3D render of the real garden, with your zones and obstacles on it.

## MORE OF WHAT YOU GET

# Tools for the days it does not just work.

Most of the year a mower needs nothing from you. Custom firmware is also for the other days: a moved server, a lost Wi-Fi password, a map that needs a nudge, a mower stuck at a hedge. Those are the days the factory firmware sent you to a support desk that no longer exists.



### Drive to a point

Tap a spot on the map and the mower drives there and stops, blades off. Only points inside a zone or on a channel are accepted. **NEXT BUILD**



### Map backup and restore

Export your complete map as one file and bring it back after a reset, a repair or on a replacement mower, then re-anchor it on the dock.



### Stays reachable when Wi-Fi fails

New router, changed password? The mower repairs a damaged Wi-Fi setting by itself, and if it still cannot get online it opens its own 'OpenNova' hotspot, so it never just disappears.



### Docks in the dark

Camera lighting for the dock approach at full brightness, so an evening mow ends on the charger instead of next to it.



### Obstacle and edge margins

Mapped obstacles stay blocked in every map the mower plans with, and you choose how far it keeps from the border.



### Recovery tools

Recalibrate the dock pose, re-anchor after a restore, reboot cleanly, clear errors, read the mower's own logs, all from the admin panel.



### SSH access

Full shell access to the mower for the curious and for remote support, so problems can be looked at instead of guessed at.



### Experimental extras

Autonomous mapping along the lawn edge and a terrain height scan from the depth camera are in, marked experimental. **BETA**

## UNDER THE HOOD

A custom build is the factory firmware image, unpacked, with Python services and boot scripts added next to the original software: an extended-command listener, a camera streamer, an LED bridge, Wi-Fi watchdogs, mDNS discovery and an SSH server. The compiled navigation, perception and motor software is untouched. That is why it can be flashed over the air like any update, and flashed back the same way.

## SIDE BY SIDE

# Factory firmware with OpenNova, or custom firmware.

Both work with OpenNova and neither needs the Novabot cloud. The left column depends on a DNS redirect on your router to keep the mower away from the cloud; the right column has no cloud address left to redirect. Everything in the right column is included from the first custom build you flash.

| FEATURE                                                       | FACTORY + OPENNOVA | CUSTOM FIRMWARE |
|---------------------------------------------------------------|--------------------|-----------------|
| Mow, pause, stop, go home                                     | ✓                  | ✓               |
| Coverage preview of the planned lanes                         | ✓                  | ✓               |
| Schedules, rain pause, notifications                          | ✓                  | ✓               |
| Mapping with the app (walk the boundary)                      | ✓                  | ✓               |
| Live position, battery, work history                          | ✓                  | ✓               |
| Works with no cloud address at all, no DNS redirect needed    | ✗                  | ✓               |
| Finds the server on its own, follows it when it moves         | ✗                  | ✓               |
| Edit zones, obstacles and channels on the map                 | ✗                  | ✓               |
| More than five zones                                          | ✗                  | ✓               |
| Edge cut                                                      | ✗                  | ✓               |
| Camera stream and headlights                                  | ✗                  | ✓               |
| Joystick with blade control                                   | ✗                  | ✓               |
| Drive to a point (next build)                                 | ✗                  | ✓               |
| Transit between zones over your channels                      | ✗                  | ✓               |
| Map backup, restore and re-anchor                             | ✗                  | ✓               |
| Stays reachable when Wi-Fi fails (own hotspot, config repair) | ✗                  | ✓               |
| Mower-side diagnostics, logs, remote support                  | ✗                  | ✓               |
| SSH access                                                    | ✗                  | ✓               |

The app itself is the same in both cases. On a factory mower the OpenNova app hides what the mower cannot do; flash custom firmware and the same app shows everything.

## WHAT IT DOES NOT DO

### Same brain, same eyes

The navigation, obstacle detection and camera AI are the original compiled Novabot software. Custom firmware adds services next to them; it does not make the mower smarter, faster or better at GPS.

### Charging station stays as it is

The charger keeps its factory firmware (0.3.6 or 0.4.0). RTK corrections, LoRa and pairing work exactly as before.

### No hardware fixes

A worn wheel motor, a wet PCB or a tired battery is still that. Firmware only changes what the software does.

### Not everything is finished

Autonomous mapping and the terrain scan are experimental. An open replacement for the mower's core software exists in the project but is not switched on in any build.

### It needs OpenNova

Custom firmware talks to your own OpenNova server on your network. Without a server there is nothing for it to talk to; the Novabot cloud is not coming back.

## GOOD TO KNOW

### Installed over the air

Flashed from the OpenNova admin panel like any firmware update. A map backup is taken first. Takes about ten minutes.

### Built on the factory firmware

Every custom build is the factory 6.0 firmware with scripts and services added. Mowers that shipped on the older 5.x line move up in the same step; no charger update needed.

### Reversible

Flash the factory firmware back through the same OTA mechanism. You lose SSH and the extras, everything else keeps working. Documented step by step.

### Your network, your rules

The mower is reachable over SSH with a default password. Change it, and keep the mower on your home network, not the open internet.

### Free and open

Built by owners, for owners, after the company behind Novabot went bankrupt. Source, builds and documentation on GitHub. No warranty, and honest release notes.

## WHO IT IS FOR

Anyone running OpenNova who wants the mower to stay useful: the owner with a big garden and six zones, the one whose router changed, the one who wants to see what the mower sees, and the one who would rather draw a new border than walk it. And anyone who wants to be able to look inside when something goes wrong.

## GETTING IT

# Ten minutes, from the admin panel.

You need an OpenNova server on your network and a mower that is online in it. Then:

- 1 Open the OpenNova admin panel and go to **Firmware**. The latest custom build is listed; the panel takes a map backup first.
- 2 Flash it. The mower downloads the build over the air, installs it and reboots. Keep it on the dock.
- 3 Check the Firmware row on the mower's page: the version now ends in **-custom**. Every custom feature in the app and the dashboard is now on.

Changed your mind? Flash the factory firmware back the same way. The step-by-step guide is in the documentation.

**Documentation**    wiki: Custom Firmware, Stock vs. Custom Firmware, Revert to stock

**Source and builds**    [github.com/rvbcrs/Novabot](https://github.com/rvbcrs/Novabot)

**Questions and bugs**    GitHub Issues, and the owner community